

# Production Order List — Detailed Design Document (詳細設計書)

002\_DD — implements 002\_BD (SCR-002), requirements REQ-020–REQ-027.

## Document control (改版履歴)

| Field             | Value                  |
|-------------------|------------------------|
| Document ID       | 002_DD                 |
| Category          | UI                     |
| System name       | ProductionManagementAI |
| Subsystem name    | Production orders      |
| Work item         | WI-003                 |
| Implements        | 002_BD                 |
| Created by        | Claude (for ThanhTN)   |
| Created date      | 2026-09-22             |
| Last updated by   | Claude (for ThanhTN)   |
| Last updated date | 2026-09-23             |

| Version | Date       | Author               | Revision content                                                                                                                                                                                                                                                               |
|---------|------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | 2026-09-22 | Claude (for ThanhTN) | Initial creation                                                                                                                                                                                                                                                               |
| 2       | 2026-09-22 | Claude (for ThanhTN) | Rendered mockup published and linked                                                                                                                                                                                                                                           |
| 3       | 2026-09-22 | Claude (for ThanhTN) | Aligned with the implementation: string-bound query parameters, enum names matched by name only, the sort applied before the projection, and the two contrast/link-style corrections axe found                                                                                 |
| 4       | 2026-09-23 | Claude (for ThanhTN) | Diagrams redrawn for the revised DD template (RFC 0009): the layout sketch and the empty/no-match states as SVG wireframes under <code>wireframes/</code> ; the screen states also drawn as a Mermaid state diagram above the table. No field, rule, API or processing changes |

| Version | Date       | Author               | Revision content                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | 2026-09-23 | Claude (for ThanhTN) | WI-005: the UI is Japanese. The message catalog lists the Japanese text with an English gloss (WI-005 DEC-003); table caption, summary, button names, formatting helpers, wireframes and mockup updated. Test-viewpoint rows keep their English shorthand for button names. A Japanese edition of the mockup was added ( mockups/002_DD_screen-b-mockup.ja.html ). No field, rule, API or processing changes |

## Overview and reference documents (概要・目次)

| Field                 | Value                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File / component name | <code>ProductionOrderListPage.tsx</code> and its child components; <code>ProductionOrdersController.List</code>                                                                 |
| Overview              | SCR-002, the production-order list: a read-only, server-paged, sortable, filterable table whose whole view state lives in the URL, and from which an order is opened in SCR-001 |

## Module / method / processing index

| No | Name                                                                      | Overview                                                                       | Notes                                |
|----|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------|
| 1  | <code>ProductionOrderListPage</code>                                      | Route component: view state, query, state selection                            | Module design §1                     |
| 2  | <code>ProductionOrderFilters</code>                                       | Filter panel, client validation, Search/Clear                                  | §2                                   |
| 3  | <code>ProductionOrderTable</code>                                         | Results table / SP cards, sorting, row activation                              | §3                                   |
| 4  | <code>ListPagination</code>                                               | Summary, page size, Previous/Next                                              | §4                                   |
| 5  | <code>ProductionOrderListQuery</code> + <code>validateFilters</code>      | The query validator — one rule set, two implementations (server authoritative) | §5                                   |
| 6  | <code>useListViewState</code>                                             | URL ⇄ typed view state                                                         | §6                                   |
| 7  | <code>ProductionOrdersController.List</code>                              | Request boundary                                                               | §7                                   |
| 8  | <code>api.ts</code> — <code>listOrders</code>                             | Typed client wrapper                                                           | §8                                   |
| 9  | <code>ProductionOrderService.ListAsync</code> and repository read methods | Server-side query use case                                                     | Pointer only — designed in 002_DD-FN |

## Reference documents

| No | Document                                  | Purpose / use                                                                                                                         | Notes      |
|----|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1  | 002_BD                                    | Screen items, validation V-09–V-13, events E-10–E-19, states                                                                          | Parent     |
| 2  | 002_DB                                    | Indexes, query shape, sort mapping, demo seed                                                                                         | Data layer |
| 3  | 002_DD-API                                | Endpoint contract                                                                                                                     | Companion  |
| 4  | 002_DD-FN                                 | Service and repository methods                                                                                                        | Companion  |
| 5  | 002_DD-SPD                                | Step-by-step processing per component                                                                                                 | Companion  |
| 6  | 001_DD, 001_DD-API, 001_DD-FN, 001_DD-SPD | SCR-001: message catalog, error contract, <code>Result&lt;T&gt;</code> , <code>ApiClient</code> , plant clock, telemetry — all reused | Screen A   |
| 7  | 0001_ADR, 0002_ADR                        | Layered backend; cookie auth and roles                                                                                                |            |
| 8  | 001_BD                                    | SCR-001's screen transition, which this screen changes (Cancel target)                                                                |            |

## Referenced by

| No | Document                                    | Purpose / use                                    | Notes           |
|----|---------------------------------------------|--------------------------------------------------|-----------------|
| 1  | 002_DD-API, 002_DD-FN, 002_DD-SPD           | Companions elaborating this DD                   |                 |
| 2  | <code>work-items/WI-003/test-plan.md</code> | Test viewpoints below become <code>TC-###</code> | Not yet written |

## Component / file organization

### X-1. Path structure

| No | Path / namespace                                                                                                   | Purpose                                                                        | Notes                                                                                                                                                                                                                                        |
|----|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <code>src/frontend/src/features/production-orders/</code>                                                          | Existing feature folder; the list page and its components join SCR-001's files | <code>ProductionOrderListPage.tsx</code> ,<br><code>ProductionOrderFilters.tsx</code> ,<br><code>ProductionOrderTable.tsx</code> ,<br><code>ListPagination.tsx</code> , <code>useListViewState.ts</code> ,<br><code>listValidation.ts</code> |
| 2  | <code>src/frontend/src/features/production-orders/api.ts</code> , <code>types.ts</code> , <code>messages.ts</code> | Existing files, extended — not duplicated                                      | <code>listOrders</code> , the list types, the new message IDs                                                                                                                                                                                |
| 3  | <code>src/backend/ProductionManagementAI.Api/Controllers/ProductionOrdersController.cs</code>                      | Existing controller, one new action                                            |                                                                                                                                                                                                                                              |

| No | Path / namespace                                                    | Purpose                                                                                           | Notes |
|----|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------|
| 4  | src/backend/ProductionManagementAI.Application/ProductionOrders/    | ProductionOrderListQuery ,<br>ProductionOrderListContracts ,<br>ListAsync on the existing service |       |
| 5  | src/backend/ProductionManagementAI.Infrastructure/ProductionOrders/ | Repository read methods, query extensions                                                         |       |
| 6  | src/backend/ProductionManagementAI.Infrastructure/Migrations/       | The two migrations from 002_DB                                                                    |       |

## X-2. Shared/common components used

| No | Name                       | Purpose                                      | Notes           |
|----|----------------------------|----------------------------------------------|-----------------|
| 1  | apiClient                  | Fetch wrapper with the 401 → /login behavior | WI-001          |
| 2  | AppHeader , ProtectedRoute | Common header and route guard                | WI-001          |
| 3  | MessageBanner              | Error/success banner with its live region    | 001_DD module 8 |
| 4  | messages.ts , statusLabels | The single message catalog and status labels | 001_DD          |

## X-3. Feature-level components used

| No | Name                   | Purpose         | Notes |
|----|------------------------|-----------------|-------|
| 1  | ProductionOrderFilters | Filter panel    | \$2   |
| 2  | ProductionOrderTable   | Results         | \$3   |
| 3  | ListPagination         | Paging controls | \$4   |

## X-4. External APIs used — see "APIs used" below.

## X-5. Responsive composition — one set of components with responsive Tailwind styling, not separate PC/SP components, except the results region:

`ProductionOrderTable` renders a `<table>` at `sm` and above and a card list below it, because a table and a card list cannot be the same markup and stay semantically correct. The filter panel is the same markup, wrapped in a `<details>`-style disclosure below `sm`.

## Companion design documents

| No | Document                                 | Type                     | Covers                                                                                                                                                                                  |
|----|------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | 002_DD-API_production-order-list.md      | api-design               | <code>GET /api/production-orders</code> (new, API-PO-04) and the reuse of <code>GET /api/products</code> ; request/response field catalogs, error codes                                 |
| 2  | 002_DD-FN_production-order-list-query.md | function-design          | <code>ProductionOrderService.ListAsync</code> , the repository's <code>CountOrdersAsync</code> / <code>ListOrdersAsync</code> , filter/sort composition, the list mapper, observability |
| 3  | 002_DD-SPD_production-order-list.md      | screen-processing-design | P-10 to P-15 plus the controller request boundary, one block per component                                                                                                              |

## Task / design index

| No | Category                 | File-level task | Function/process-level task               | Item-level task               | Confirmed | Issue | Reviewer | Reworked | Date       | Notes                                                                    |
|----|--------------------------|-----------------|-------------------------------------------|-------------------------------|-----------|-------|----------|----------|------------|--------------------------------------------------------------------------|
| 1  | basic-design             | 002_BD          | SCR-002                                   | —                             | yes       | —     | ThanhTN  | no       | 2026-09-22 | Approved ("the BD look good"); revised to v2/v3 during 002_DB and 002_DD |
| 2  | database-design          | 002_DB          | Indexes, seed, queries                    | —                             | yes       | —     | ThanhTN  | no       | 2026-09-22 | Approved ("the DB design is approved")                                   |
| 3  | detailed-design          | 002_DD          | Modules §1–§8, states, mockup             | —                             | no        | —     | ThanhTN  | no       | 2026-09-22 | This document; mockup published 2026-09-22                               |
| 4  | api-design               | 002_DD-API      | <code>GET /api/production-orders</code>   | Query parameters, error codes | no        | —     | ThanhTN  | no       | 2026-09-22 |                                                                          |
| 5  | function-design          | 002_DD-FN       | <code>ListAsync</code> , repository reads | Sort mapping, overdue rule    | no        | —     | ThanhTN  | no       | 2026-09-22 |                                                                          |
| 6  | screen-processing-design | 002_DD-SPD      | P-10–P-15                                 | Response handling             | no        | —     | ThanhTN  | no       | 2026-09-22 |                                                                          |

# Module design

## 1. ProductionOrderListPage

| Field                   | Value                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Description             | Route component for <code>/production-orders</code> : owns the query lifecycle and chooses which of the six screen states to render |
| Return type             | JSX element                                                                                                                         |
| Created by / date       | Claude / 2026-09-22                                                                                                                 |
| Last modified by / date | —                                                                                                                                   |

Preconditions: rendered inside `ProtectedRoute` , so a session exists.

Not applicable — plain component, no rule/validator fields.

**Arguments:** none (route component).

### Dependencies

| No | Module / component                                                                                                                 | Overview                                         | Notes  |
|----|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------|
| 1  | <code>useListViewState</code>                                                                                                      | View state from the URL                          | §6     |
| 2  | <code>api.ts</code> <code>listOrders</code> , <code>listProducts</code>                                                            | Data                                             | §8     |
| 3  | <code>ProductionOrderFilters</code> , <code>ProductionOrderTable</code> , <code>ListPagination</code> , <code>MessageBanner</code> | Children                                         | §2–§4  |
| 4  | <code>useAuth</code>                                                                                                               | Role gate (UX only; the server is authoritative) | WI-001 |

Processing overview: see 002\_DD-SPD §1 and §6. One query per distinct view state; a superseded response is discarded rather than rendered.

**Processing flow:** 002\_DD-SPD §1 (P-10) and §6 (P-15).

### Return value

| Type | Name | Description                                                                                                                                |
|------|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| JSX  | —    | One of <code>loading</code> , <code>ready</code> , <code>empty</code> , <code>noMatch</code> , <code>error</code> , <code>forbidden</code> |

## 2. ProductionOrderFilters

| Field                   | Value                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------|
| Description             | The filter panel (items 7–13): holds draft filter values and publishes them on Search |
| Return type             | JSX element                                                                           |
| Created by / date       | Claude / 2026-09-22                                                                   |
| Last modified by / date | —                                                                                     |

Preconditions: the product options have loaded, or the product select is rendered disabled with its load error.

Not applicable — plain component; its rules live in §5.

### Arguments

| No | Type                              | Name                  | Description                                                |
|----|-----------------------------------|-----------------------|------------------------------------------------------------|
| 1  | <code>ListViewState</code>        | <code>view</code>     | Current applied filters, used to (re-)initialize the draft |
| 2  | <code>ProductResponse[]</code>    | <code>products</code> | Options for item 8                                         |
| 3  | <code>(filters) =&gt; void</code> | <code>onApply</code>  | Publishes valid filters to the view state                  |
| 4  | <code>boolean</code>              | <code>busy</code>     | Disables Search while a query is in flight                 |

### Dependencies

| No | Module / component           | Overview               | Notes |
|----|------------------------------|------------------------|-------|
| 1  | <code>validateFilters</code> | Client-side V-09, V-10 | §5    |
| 2  | <code>messages.ts</code>     | ID → text              |       |

Processing overview: 002\_DD-SPD §3 (P-12). The draft re-bases whenever the applied view state changes, so Back or a shared URL always leaves the panel showing what produced the rows.

**Return value:** JSX (a `<form>`).

### 3. ProductionOrderTable

| Field                   | Value                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------|
| Description             | The results region (items 16–22, 27): a table at <code>sm</code> and above, a card list below it |
| Return type             | JSX element                                                                                      |
| Created by / date       | Claude / 2026-09-22                                                                              |
| Last modified by / date | —                                                                                                |

Preconditions: called only in the `ready` state, with `items.length ≥ 0`.

Not applicable — plain component.

#### Arguments

| No | Type                                   | Name                | Description                            |
|----|----------------------------------------|---------------------|----------------------------------------|
| 1  | <code>ProductionOrderListItem[]</code> | <code>items</code>  | The page's rows                        |
| 2  | <code>ProductionOrderSort</code>       | <code>sort</code>   | Active sort key, from the response     |
| 3  | <code>'asc' \  'desc'</code>           | <code>dir</code>    | Active direction, from the response    |
| 4  | <code>(sort, dir) =&gt; void</code>    | <code>onSort</code> | Publishes a new sort to the view state |

#### Dependencies

| No | Module / component                                                           | Overview                    | Notes                                                              |
|----|------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------|
| 1  | <code>Link</code>                                                            | Order-number cell → SCR-001 | react-router                                                       |
| 2  | <code>statusLabels</code> , <code>labels</code> , date and number formatting | M-05, M-10                  | <code>messages.ts</code> ; <code>src/lib/format.ts</code> (WI-005) |

Processing overview: 002\_DD-SPD §4 (P-13). Accessibility specifics: `<caption>` naming the table and its sort, 「製造指示一覧（納期の昇順）」 (Production orders, sorted by due date ascending), `<th scope="col">` header cells each containing a `<button>`, `aria-sort="ascending" | "descending"` on the active header only, the overdue marker as text next to the due date, and the row's mouse handler deliberately additive to the link.

**Return value:** JSX.



## 4. ListPagination

| Field                   | Value                                                                  |
|-------------------------|------------------------------------------------------------------------|
| Description             | Result summary (item 14), page-size select (15) and Previous/Next (23) |
| Return type             | JSX element                                                            |
| Created by / date       | Claude / 2026-09-22                                                    |
| Last modified by / date | —                                                                      |

Preconditions: a response with `total`, `page` and `pageSize` exists.

Not applicable — plain component.

### Arguments

| No | Type                                                            | Name                                                           | Description                        |
|----|-----------------------------------------------------------------|----------------------------------------------------------------|------------------------------------|
| 1  | <code>number</code>                                             | <code>total</code> , <code>page</code> , <code>pageSize</code> | From the response, not the request |
| 2  | <code>(page) =&gt; void</code> / <code>(size) =&gt; void</code> | <code>onPage</code> , <code>onPageSize</code>                  | Publish to the view state          |

**Dependencies:** none beyond §6.

Processing overview: 002\_DD-SPD §5 (P-14). The summary is the screen's live region: when a query finishes, it announces 「{total}件中 {first}～{last}件」 ({first}–{last} of {total} orders) (M-09).

**Return value:** JSX.

## 5. ProductionOrderListQuery (server) and validateFilters (client)

| Field                   | Value                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Description             | The list query's validator and normalizer: one rule set stated once here, implemented on both sides, with the server authoritative |
| Return type             | <code>Result&lt;ProductionOrderListQuery&gt;</code> (server) / <code>Record&lt;field, messageId&gt;</code> (client)                |
| Created by / date       | Claude / 2026-09-22                                                                                                                |
| Last modified by / date | —                                                                                                                                  |

Preconditions: none — it is a pure function of the raw query values.

Rule type: input validation plus normalization (format, range, cross-field and allow-list checks).

### Rule references

| No | Rule / validator                | Location            | Notes                                                                      |
|----|---------------------------------|---------------------|----------------------------------------------------------------------------|
| 1  | 002_BD V-09–V-13                | 002_BD §5           | The rules this implements                                                  |
| 2  | <code>ProductExistsAsync</code> | 002_DD-FN §1 step 2 | The one check that needs the database, so it runs in the service, not here |

### Condition / data references

| No | Key                            | Source                                                              | Notes                    |
|----|--------------------------------|---------------------------------------------------------------------|--------------------------|
| 1  | Allowed statuses               | <code>ProductionOrderStatus</code> enum                             | Domain                   |
| 2  | Allowed sort keys / directions | <code>ProductionOrderSort</code> , <code>SortDirection</code> enums | Defined with this module |
| 3  | Allowed page sizes             | <code>{10, 20, 50, 100}</code> constant                             | DEC-002                  |

### Check parameters

| No | Name                                                                             | Type          | Target field                                    |
|----|----------------------------------------------------------------------------------|---------------|-------------------------------------------------|
| 1  | <code>status</code>                                                              | string[]      | Status filter (V-11)                            |
| 2  | <code>productId</code>                                                           | string        | Product filter (V-12, existence in the service) |
| 3  | <code>dueFrom</code> , <code>dueTo</code>                                        | string (date) | Due-date range (V-10)                           |
| 4  | <code>orderNumber</code>                                                         | string        | Fragment (V-09)                                 |
| 5  | <code>sort</code> , <code>dir</code> , <code>page</code> , <code>pageSize</code> | string / int  | Sort and paging (V-13)                          |

### Arguments

| No | Type             | Name                 | Description                                                  |
|----|------------------|----------------------|--------------------------------------------------------------|
| 1  | raw query values | <code>request</code> | Query-string values (server) or draft filter values (client) |

**Dependencies:** none — deliberately pure, so both sides can unit-test it without a database or a DOM.

Processing overview: validate, then normalize. Absent values take defaults; present-but-invalid values are rejected with their message ID and never silently defaulted (002\_BD V-13) — that asymmetry is the whole point of the module, because defaulting a bad value would let a crafted request widen the result set. Normalization

covers the order-number fragment (trim → length check → upper-case → escape \ , % , \_ → wrap in %...%) and de-duplicating the status set. Every offending parameter is reported in one response, not the first one only.

The client implements V-09 and V-10 only (the two the user can type wrong); V-11, V-12 and V-13 cannot be violated through the UI, since those values come from the screen's own controls, so the client does not re-implement them — the server still enforces all five.

**Processing flow:** 002\_DD-SPD §7 step 3 (server) and §3 step 3 (client).

**Return value**

| Type                             | Name   | Description                                    |
|----------------------------------|--------|------------------------------------------------|
| Result<ProductionOrderListQuery> | server | Ok(query) or Invalid(parameter → message ID)   |
| Record<string, string>           | client | Empty when valid; field → message ID otherwise |

**6. useListViewState**

| Field                   | Value                                                                 |
|-------------------------|-----------------------------------------------------------------------|
| Description             | Hook converting between the URL query string and the typed view state |
| Return type             | [ListViewState, (next, options) => void]                              |
| Created by / date       | Claude / 2026-09-22                                                   |
| Last modified by / date | —                                                                     |

Preconditions: rendered inside the router.

Not applicable — plain hook.

**Arguments:** none.

**Dependencies**

| No | Module / component | Overview   | Notes        |
|----|--------------------|------------|--------------|
| 1  | useSearchParams    | URL access | react-router |

Processing overview: 002\_DD-SPD §2 (P-11). Parsing is total and never throws; serialization omits every parameter equal to its default.

## Return value

| Type                     | Name | Description                                                |
|--------------------------|------|------------------------------------------------------------|
| [ListViewState, setView] | —    | Current view state and the publisher that rewrites the URL |

## 7. ProductionOrdersController.List

| Field                   | Value                                              |
|-------------------------|----------------------------------------------------|
| Description             | GET /api/production-orders — the request boundary  |
| Return type             | ActionResult<PagedResult<ProductionOrderListItem>> |
| Created by / date       | Claude / 2026-09-22                                |
| Last modified by / date | —                                                  |

Preconditions: the existing `ProductionOrderEditor` policy has authorized the request.

Not applicable — plain handler.

## Arguments

| No | Type                       | Name              | Description                                |
|----|----------------------------|-------------------|--------------------------------------------|
| 1  | ProductionOrderListRequest | request           | [FromQuery] binding of the nine parameters |
| 2  | CancellationToken          | cancellationToken |                                            |

## Dependencies

| No | Module / component                     | Overview         | Notes         |
|----|----------------------------------------|------------------|---------------|
| 1  | ProductionOrderListQuery.TryCreate     | Validation       | \$5           |
| 2  | ProductionOrderService.ListAsync       | Use case         | 002_DD-FN \$1 |
| 3  | ProductionOrderProblems.ToActionResult | Result<T> → HTTP | Existing      |

Processing overview: 002\_DD-SPD §7. No `[Consumes]` attribute — the endpoint takes no body.

**Return value:** 200 with the paged result, or the Problem Details responses catalogued in 002\_DD-API.

## 8. `api.ts` — `listOrders`

| Field                   | Value                                                                  |
|-------------------------|------------------------------------------------------------------------|
| Description             | Typed wrapper building the query string and parsing the paged response |
| Return type             | <code>Promise&lt;PagedResult&lt;ProductionOrderListItem&gt;&gt;</code> |
| Created by / date       | Claude / 2026-09-22                                                    |
| Last modified by / date | —                                                                      |

Preconditions: none.

Not applicable — plain function.

### Arguments

| No | Type                       | Name                | Description                                                                                              |
|----|----------------------------|---------------------|----------------------------------------------------------------------------------------------------------|
| 1  | <code>ListViewState</code> | <code>view</code>   | Serialized with the same rules as the URL (§6), so what the user sees and what is queried cannot diverge |
| 2  | <code>AbortSignal</code>   | <code>signal</code> | Cancels a superseded query                                                                               |

**Dependencies:** `ApiClient` (401 handling, JSON parsing, Problem Details mapping).

Processing overview: one serializer for both the URL and the request; defaults are omitted from both.

**Return value:** the parsed `PagedResult`, or a typed API error the page maps per 002\_DD-SPD §6.

## 9. Server-side service and repository methods

Pointer only — `ProductionOrderService.ListAsync`, `IProductionOrderRepository.CountOrdersAsync` / `ListOrdersAsync`, the filter/sort composition and the list mapper are designed in **002\_DD-FN**, which also specifies the spans and metrics for the endpoint.

## Screen layout and mockup

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Implementation-fidelity layout. It refines 002\_BD §1 in two ways: the filter panel is laid out as a labelled grid that reflows from four columns to one, and the overdue marker is specified as a text badge after the due date rather than a styling of the date itself, so it survives both themes and screen readers.

1 ProductionManagementAI

ダッシュボード

製造指示一覧

+ 新規製造指示

2 Admin · ログアウト

3 ホーム > 製造指示一覧

4 製造指示一覧

5 + 新規製造指示

6 Only in the error state: 「問題が発生しました。もう一度お試しください。」

再試行

絞り込み

7 ステータス

☒ 下書き

☒ 進行中

☐ 完了

☐ 取消

8 製品

すべての製品

9 納期（開始）

2026/09/01

error text slot

10 納期（終了）

2026/10/31

error text slot

11 指示番号

00042

error text slot

12 クリア

13 検索

14 87件中 1～20件

15 表示件数

20

| 16 指示番号 ▲        | 17 製品              | 18 数量 | 19 納期      | 20 ステータス    | 21 更新日時          |
|------------------|--------------------|-------|------------|-------------|------------------|
| 22 PO-2026-00042 | P-1004 — ドライブシャフト  | 250   | 2026/09/18 | 納期遅れ 27 進行中 | 2026/09/20 10:12 |
| PO-2026-00043    | P-1001 — ブレーキキャリパー | 120   | 2026/09/30 | 下書き         | 2026/09/19 14:03 |
| ...              |                    |       |            |             |                  |

23 < 前へ

1 / 5 ページ

次へ >

Filters reflow from a four-column grid to one column below 640px. The overdue marker is a text badge after the due date.

Empty state (no orders at all) and no-match state (filters set, nothing matched):

24

Empty state — no orders at all

製造指示はまだありません。最初の1件を登録してください。

+ 新規製造指示

Replaces items 14–23.

25

No-match state — filters set, nothing matched

絞り込みに一致する製造指示はありません。  
納期の範囲を広げるか、製品の絞り込みを外してください。

絞り込みをクリア

The filter panel above keeps its values.

Mockup artifact: <https://claude.ai/artifact/2XrZ9xnEfzQ6pCbUnZovL3> — published privately on 2026-09-22 with the user's authorization (plan revision 1, "Resources and external actions"). Source kept in the repository at `docs/en/020_detailed-design/002/mockups/002_DD_screen-b-mockup.html` , alongside 001\_DD's, whose token system and app-look CSS it reuses so the two mockups read as one set. Seven artboards: 1 default view, 2 filtered and sorted (with the focus rings), 3 filter validation errors, 4 no-match, 5 empty, 6 query error, 7 SP card layout.

Japanese edition, with the page captions in Japanese too, linked from the Japanese PDF (WI-005): <https://claude.ai/artifact/TTxzX1PJVyKr3aLYpHSRZK> (private). Source: [mockups/002\\_DD\\_screen-b-mockup.ja.html](mockups/002_DD_screen-b-mockup.ja.html) .

| Region           | Contains (field/control)              | Notes                                                                                                                                                    |
|------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Header           | AppHeader (items 1, 2)                | Shared component, unchanged                                                                                                                              |
| Title row        | Heading (4), New production order (5) | Button right-aligned on PC, full width on SP                                                                                                             |
| Banner           | MessageBanner (6)                     | Rendered only in the <code>error</code> state; <code>role="alert"</code> , carries 再試行 (Retry)                                                           |
| Filter panel     | Items 7–13                            | 4-column grid at <code>lg</code> , 2 at <code>sm</code> , 1 below; each field has a reserved error slot so the layout doesn't jump when an error appears |
| Summary row      | Items 14, 15                          | Summary is the live region ( <code>role="status"</code> )                                                                                                |
| Results          | Items 16–22, 27                       | <code>&lt;table&gt;</code> at <code>sm</code> + , card list below; sticky header row on PC                                                               |
| Paging           | Item 23                               | Centered; controls disabled, not hidden, at the ends                                                                                                     |
| Empty / no-match | Items 24, 25                          | Replace the results region only; the filter panel stays                                                                                                  |



# Screen item definition

Types and client-side behavior for each input; the authoritative rules are 002\_BD §5 and the server (§5 above).

| Field       | Type                                                    | Required | Validation rule                                                                                                                                                     | Source (BD ref)           |
|-------------|---------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| status      | Set<'Draft'\  'InProgress'\  'Completed'\  'Cancelled'> | no       | Four checkboxes in a <fieldset> with a <legend>; empty set = no restriction                                                                                         | §3 item 7, V-11, DEC-005  |
| productId   | string \  null (uuid)                                   | no       | Native <select>, options ordered by SKU, label "{sku} — {name}" (e.g. P-1004 – ドライブシャフト), first option 「すべての製品」 (All products) with an empty value                  | §3 item 8, V-12, M-06     |
| dueFrom     | string \  null (YYYY-MM-DD)                             | no       | <input type="date">; malformed → MSG-E016; after dueTo → MSG-E017                                                                                                   | §3 item 9, V-10           |
| dueTo       | string \  null (YYYY-MM-DD)                             | no       | <input type="date">; malformed → MSG-E016                                                                                                                           | §3 item 10, V-10          |
| orderNumber | string \  null                                          | no       | <input type="search" maxlength="20">; trimmed on submit; longer than 20 → MSG-E015 (the attribute prevents it in practice, the check covers paste and crafted URLs) | §3 item 11, V-09          |
| sort        | ProductionOrderSort                                     | no       | Set by the header buttons only; never typed                                                                                                                         | §3 items 16–21, V-13      |
| dir         | 'asc' \  'desc'                                         | no       | Toggled by re-clicking the active header                                                                                                                            | §3 items 16–21, V-13      |
| page        | number                                                  | no       | Set by the paging controls; reset to 1 by any filter, sort or page-size change                                                                                      | §3 item 23, V-13          |
| pageSize    | 10 \  20 \  50 \  100                                   | no       | <select>, default 20                                                                                                                                                | §3 item 15, V-13, DEC-002 |

Read-only row fields (items 16–22, 27) are rendered from ProductionOrderListItem (002\_DD-API) and have no input behavior. isOverdue comes from the server, so every client agrees on "today" (WI-002 DEC-011).

Message catalog additions (messages.ts; the server returns the same IDs as code; other UI strings are in its labels object, WI-005 DEC-007). 001\_DD already defines MSG-E001–MSG-E014 and MSG-I001–MSG-I002, so Screen B continues the same catalog rather than starting a second one:

| ID       | Text (Japanese, as implemented) | English gloss                                   |
|----------|---------------------------------|-------------------------------------------------|
| MSG-E015 | 指示番号の検索は20文字以内で入力してください。        | Order number search can't exceed 20 characters. |

| ID       | Text (Japanese, as implemented) | English gloss                                        |
|----------|---------------------------------|------------------------------------------------------|
| MSG-E016 | 有効な日付を入力してください。                 | Enter a valid date.                                  |
| MSG-E017 | 納期（開始）は納期（終了）以前の日付にしてください。      | 'Due from' must be on or before 'Due to'.            |
| MSG-E018 | 不明なステータスです。                     | Unknown order status.                                |
| MSG-E019 | 対応していない並べ替えまたはページングの指定です。       | Unsupported sort or paging option.                   |
| MSG-E020 | 製造指示を閲覧する権限がありません。              | You don't have permission to view production orders. |
| MSG-I003 | 製造指示はまだありません。最初の1件を登録してください。    | No production orders yet. Create the first one.      |
| MSG-I004 | 絞り込みに一致する製造指示はありません。            | No orders match your filters.                        |

Reused unchanged: MSG-E002 (unknown product filter), MSG-E013 (query failure).

Two visual rules were corrected during implementation, after `@axe-core/playwright` failed on the real screen: the `Cancelled` badge uses `text-gray-600` rather than `text-gray-400` (2.9:1 on white fails WCAG 1.4.3), and the breadcrumb link is underlined always, not only on hover (WCAG 1.4.1, "link in text block").

## Loading / empty / error / success states

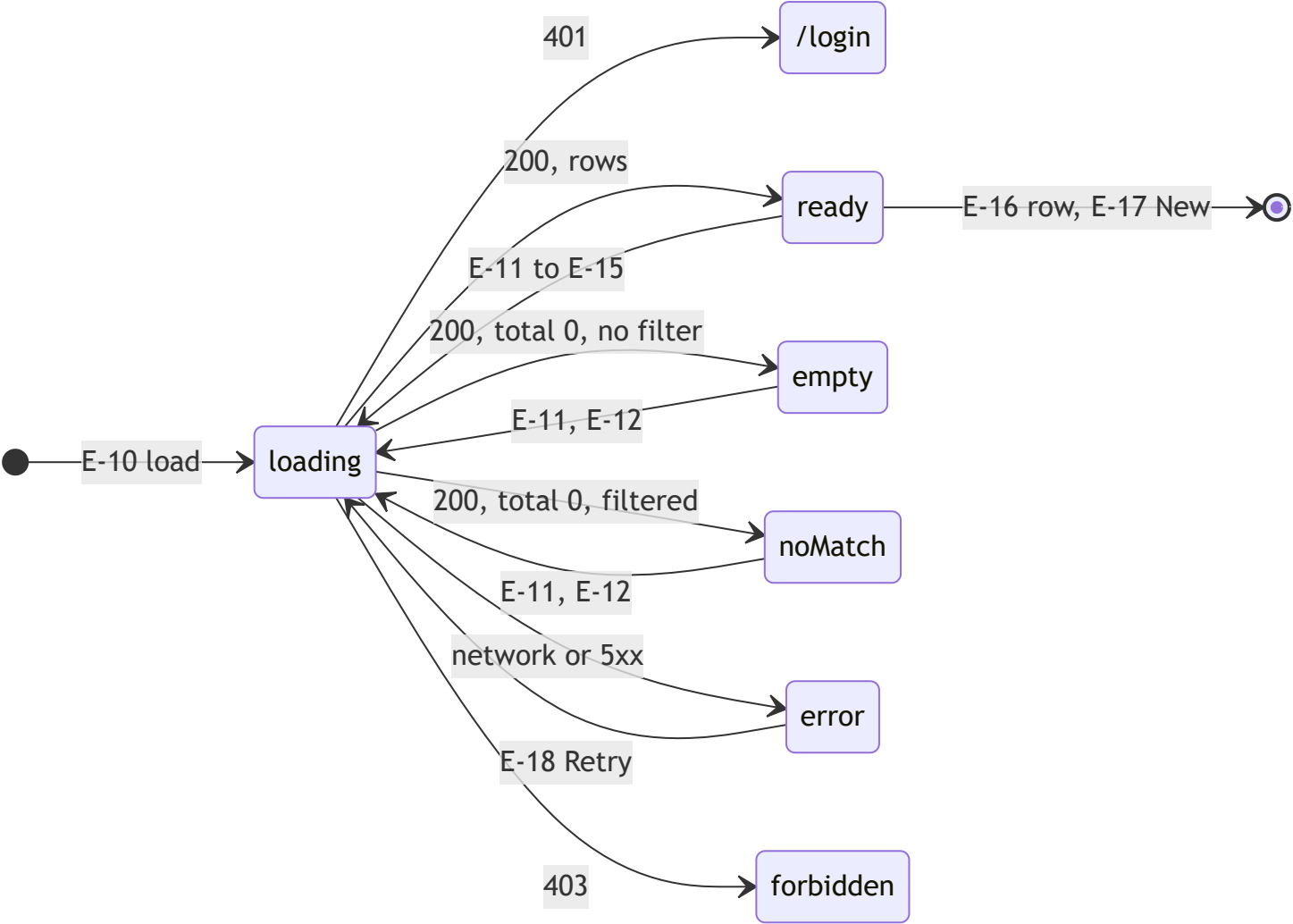
| State                                   | Trigger                                                        | UI behavior                                                                                                                                                                                                | Data shown      |
|-----------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <code>loading</code>                    | A query is in flight                                           | The results region is replaced by a skeleton with <code>aria-busy="true"</code> ; the filter panel stays usable but 検索 (Search) is disabled; previous rows are not left visible as if they were the answer | Skeleton rows   |
| <code>ready</code>                      | 200 with <code>total &gt; 0</code>                             | Table, summary, sort marker and paging, all rendered from the response's echoed controls; the summary's live region announces the new total                                                                | The page's rows |
| <code>ready</code> (past the last page) | 200 with <code>total &gt; 0</code> and <code>items = []</code> | Empty table body with the paging controls intact and < 前へ (Previous) enabled — not an error (REQ-024)                                                                                                      | No rows         |
| <code>empty</code>                      | 200 with <code>total = 0</code> and no filter set              | MSG-I003 with a + 新規製造指示 (New production order) action, replacing the results region                                                                                                                       | —               |
| <code>noMatch</code>                    | 200 with <code>total = 0</code> and at least one filter set    | MSG-I004 with 絞り込みをクリア (Clear filters); the filter panel keeps its values so the user can adjust rather than retype                                                                                        | —               |
| Filter validation error                 | Client V-09/V-10, or a 400 from the server                     | Inline error under each offending field, focus moved to the first one; the previously loaded rows stay on screen and are not implied to match the rejected filters                                         | Previous rows   |

| State     | Trigger                             | UI behavior                                                              | Data shown            |
|-----------|-------------------------------------|--------------------------------------------------------------------------|-----------------------|
| error     | Network failure or 5xx              | MSG-E013 banner with 再試行 (Retry), which re-runs the same query unchanged | Previous rows cleared |
| forbidden | Client role gate or 403             | Full-region panel MSG-E020 with a link back to / ; no query is issued    | —                     |
| 401       | Any request without a valid session | apiClient redirects to /login                                            | —                     |

## Processing and state transitions

This screen has no business state machine — it changes no order data (DEC-003). Its transitions are view-state transitions.

State transitions



Transitions from any state (E-17 New, E-19 Back/Forward) and a 400 that keeps the previous state are in the table below, not the diagram.

| From state              | Event                      | To state                      | Side effect                                           |
|-------------------------|----------------------------|-------------------------------|-------------------------------------------------------|
| (route entered)         | E-10 load                  | loading                       | View state read from the URL; product options fetched |
| loading                 | 200, rows                  | ready                         | Summary announced                                     |
| loading                 | 200, total = 0 , no filter | empty                         | —                                                     |
| loading                 | 200, total = 0 , filtered  | noMatch                       | —                                                     |
| loading                 | 400                        | previous state + field errors | No rows replaced                                      |
| loading                 | 401 / 403                  | /login / forbidden            | —                                                     |
| loading                 | network or 5xx             | error                         | —                                                     |
| ready / empty / noMatch | E-11 Search, E-12 Clear    | loading                       | URL updated (push), page → 1                          |
| ready                   | E-13 sort                  | loading                       | URL updated (push), page → 1                          |
| ready                   | E-14 page, E-15 page size  | loading                       | URL updated (push); page size also resets page → 1    |
| ready                   | E-16 row activated         | (leaves screen)               | Navigates to SCR-001 edit mode                        |
| any                     | E-17 New                   | (leaves screen)               | Navigates to SCR-001 create mode                      |
| error                   | E-18 Retry                 | loading                       | Same query re-run                                     |
| any                     | E-19 Back/Forward          | loading                       | View state re-read from the URL                       |

## Processing flows

| Flow                            | Where (screen-processing-design / function-design section) | Events          |
|---------------------------------|------------------------------------------------------------|-----------------|
| P-10 Load and query             | 002_DD-SPD §1                                              | E-10, E-19      |
| P-11 View-state synchronization | 002_DD-SPD §2                                              | E-10–E-15, E-19 |
| P-12 Filter, search, clear      | 002_DD-SPD §3                                              | E-11, E-12      |
| P-13 Sort and row activation    | 002_DD-SPD §4                                              | E-13, E-16      |
| P-14 Page and page size         | 002_DD-SPD §5                                              | E-14, E-15      |
| P-15 Query-response handling    | 002_DD-SPD §6                                              | E-10, E-18      |
| Server query use case           | 002_DD-FN §1–§6                                            | —               |

## APIs used

| Endpoint               | Method | Purpose                           | Design doc                               |
|------------------------|--------|-----------------------------------|------------------------------------------|
| /api/production-orders | GET    | One page of orders with the total | 002_DD-API §1                            |
| /api/products          | GET    | Product filter options            | 002_DD-API §2 (defined in 001_DD-API §1) |

## Database and transaction mapping

| Operation                      | Table(s)                             | Transaction boundary                    | Concurrency handling       |
|--------------------------------|--------------------------------------|-----------------------------------------|----------------------------|
| Count matches                  | production_orders                    | None (single statement, READ COMMITTED) | Not applicable — read-only |
| Fetch page                     | production_orders joined to products | None (single statement)                 | Not applicable — read-only |
| Product filter existence check | products                             | None                                    | Not applicable             |

Deliberate exception to the one-query-per-table convention: the page query joins `products`, because the list must show the product label and can sort by it, and issuing a second query per page to resolve 20 product ids would be slower and could tear (a product renamed between the two queries). The count query does **not** join, since no filter needs it. Both select only the columns the screen uses (002\_DB read projection). No existing SCR-001 query is modified.

The two statements can observe different snapshots, so the total may differ from the page by a row created between them; 002\_DB accepts that explicitly rather than taking a repeatable-read transaction for an informational screen.

## Exception handling

| Failure                       | Retry policy                          | Timeout | User-facing error                                             | Logging                                                                   |
|-------------------------------|---------------------------------------|---------|---------------------------------------------------------------|---------------------------------------------------------------------------|
| Invalid query parameter (400) | none automatic — the user corrects it | —       | Inline field error, or a banner for a parameter with no field | Information: rejected parameter names with their message IDs (structured) |
| Unknown product filter (400)  | none                                  | —       | Inline error MSG-E002 on the product field                    | Information                                                               |
| Not signed in (401)           | none                                  | —       | Redirect to <code>/login</code>                               | Existing <code>apiClient</code> behavior                                  |
| Missing role (403)            | none                                  | —       | Panel MSG-E020                                                | Information                                                               |

| Failure                                             | Retry policy                           | Timeout                              | User-facing error                                                                     | Logging                                                                                                                                       |
|-----------------------------------------------------|----------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Query/database failure (500)                        | none automatic; 再試行 (Retry) re-runs it | Npgsql command timeout 30s (default) | Banner MSG-E013                                                                       | Error with exception, server-side only; the response is generic Problem Details with no stack trace or exception text ( ai/rules/backend.md ) |
| Frontend network failure                            | none automatic; 再試行 (Retry)            | browser default                      | Banner MSG-E013                                                                       | <code>console.error</code> in development only                                                                                                |
| Superseded query (user changed the view mid-flight) | not an error                           | —                                    | none                                                                                  | none — the response is discarded by <code>AbortSignal</code>                                                                                  |
| Product options fail to load                        | none automatic                         | —                                    | Product select disabled with an inline load error; the rest of the screen still works | Information                                                                                                                                   |

## Test viewpoints and unresolved decisions

U = unit, I = integration, E = E2E. These become `TC-###` when the WI-003 test plan is written (plan revision 2).

| Scenario                                             | Precondition     | Expected result                                                                                                                                                                                                                | Test-plan ID |
|------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Default view (REQ-020) — I, E                        | Seeded 80 orders | 200; 20 rows; <code>total = 80</code> ; sorted by due date ascending; every status present (DEC-006)                                                                                                                           | TC-101       |
| Row content (REQ-021) — U, E                         | One known order  | Row shows order number, SKU + name, quantity, due date, status label, updated date; no GUID and no raw enum name visible                                                                                                       | TC-102       |
| Overdue marker (REQ-021) — U                         | Plant today = D  | <code>dueDate &lt; D</code> with <code>Draft / InProgress</code> → <code>isOverdue = true</code> ; same date with <code>Completed / Cancelled</code> → <code>false</code> ; <code>dueDate = D</code> → <code>false</code>      | TC-103       |
| Status filter, multi-select (REQ-022) — I            | Seeded data      | <code>?status=Draft&amp;status=InProgress</code> returns exactly those rows and a matching total; no <code>status</code> returns everything                                                                                    | TC-104       |
| Product / due-range / fragment filters (REQ-022) — I | Seeded data      | Each filter alone returns exactly the matching rows; range inclusive at both ends; fragment matches case-insensitively mid-string                                                                                              | TC-105       |
| Combined filters (REQ-022) — I                       | Seeded data      | Filters combine with AND; the total matches the filtered row count                                                                                                                                                             | TC-106       |
| Invalid filter values (REQ-022) — I                  | —                | Unknown status → 400 MSG-E018; unknown product → 400 MSG-E002; <code>dueFrom &gt; dueTo</code> → 400 MSG-E017; malformed date → 400 MSG-E016; 21-character fragment → 400 MSG-E015; all offending parameters reported together | TC-107       |

| Scenario                                         | Precondition                               | Expected result                                                                                                                                                                                                                                       | Test-plan ID |
|--------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Fragment escaping (REQ-022, DEC-010) — I         | An order <code>P0-2026-00042</code> exists | Searching <code>%</code> or <code>_</code> returns no rows (treated literally), not every row                                                                                                                                                         | TC-108       |
| Sorting (REQ-023) — I, E                         | Seeded data                                | Each of the six keys sorts the whole result set, not just the page; clicking again reverses; status sorts in workflow order; unsupported key → 400 MSG-E019                                                                                           | TC-109       |
| Paging stability (REQ-023, REQ-024) — I          | Orders sharing a due date                  | Pages 1 and 2 together contain each row exactly once (the order-number tie-breaker)                                                                                                                                                                   | TC-110       |
| Paging and page size (REQ-024) — I, E            | 80 orders                                  | <code>pageSize</code> 10/20/50/100 accepted, default 20; page beyond the last → 200 with <code>items = []</code> and the real <code>total</code> ; <code>pageSize=101</code> , <code>0</code> , <code>abc</code> , <code>page=0</code> → 400 MSG-E019 | TC-111       |
| Page reset (REQ-024) — U, E                      | On page 3                                  | Changing a filter, the sort or the page size returns to page 1                                                                                                                                                                                        | TC-112       |
| Navigation (REQ-025) — E                         | Rows present                               | Clicking a row, and activating the order-number link by keyboard, both open <code>/production-orders/{id}</code> ; <b>New production order</b> opens <code>/production-orders/new</code> ; the list exposes no save or status control                 | TC-113       |
| Authorization (REQ-026) — I, E                   | —                                          | Unauthenticated → 401 and the UI redirects to <code>/login</code> ; signed in without either role → 403 and no order data in the response body                                                                                                        | TC-114       |
| URL view state (REQ-027) — E                     | A filtered, sorted, paged view             | Reload reproduces it; Back returns to the previous view; a URL with <code>sort=bogus&amp;page=-1</code> renders the default view without an error page                                                                                                | TC-115       |
| Empty and no-match states (REQ-020, REQ-022) — E | Empty table / non-matching filter          | MSG-I003 with <b>New production order</b> / MSG-I004 with <b>Clear filters</b> , and the filter panel keeps its values                                                                                                                                | TC-116       |
| Query failure (REQ-020) — E                      | API returns 500                            | MSG-E013 banner with <b>Retry</b> ; <b>Retry</b> re-runs the same query                                                                                                                                                                               | TC-117       |
| Accessibility — U (axe), E (axe)                 | Table rendered                             | No axe violations; <code>aria-sort</code> on the active column only; table has a caption; every row reachable by keyboard; summary announces the total                                                                                                | TC-118       |
| Index usage (002_DB) — I                         | Seeded data                                | The default-sort and fragment queries use <code>ix_production_orders_due_date_order_number</code> and the trigram index ( <code>EXPLAIN</code> ), not a sequential scan                                                                               | TC-119       |

Unresolved decisions: none for the design. Two items are recorded as limitations rather than decisions:

- The list endpoint reuses the `ProductionOrderEditor` policy, so a read-only role cannot be expressed. This waits on WI-001 DEC-015 (the full role/permission matrix), which remains open.
- The mockup is a static rendering, not a prototype: it fixes layout, state coverage and the treatment of status, the overdue marker and the sort indicator, but no interaction is demonstrated there — behavior is specified in 002\_DD-SPD.